



enemies, the weapons, the ammunition and the explosions. These .WAD files were easy to modify, create and distribute over the Internet. The Modscene started with this game, and within a span of two years thousands of .WAD files were created and distributed over the Internet. More than one thousand five hundred of these .WAD files were distributed during later versions of the Doom game, and the game enjoyed a market life-time of more than five years because the modifications required the original game to be installed on the system. Another important feature about the modification community around Doom was that the developers released tools specifically meant for modding. Id Software encouraged the modification community to the extent, that they even released the source code of the tools used to create the game. A number of map editing utilities by third party developers also came out. Another important aspect established by *Doom* was that the copyright of the game was limited to the library, but the game engine was released with a more modification friendly license. In 1997, the source code of the game was released with a GNU licence, paving the way for gamers to have more control over their modifications. The producers of games realised at this time that allowing the gaming community to modify their games boosted their sales. Valve and Id Software in particular are known to encourage the

Modscene actively, to the extent of hiring the biggest contributors to the modscene. A number of people who made .WAD files played an active role in developing *Quake*, *Unreal Tournament* and *Half Life*. All these games are known to have active modding communities. A few of these mods are actually played more than the original games.

Rocket science

One of the reasons for the popularity of *Doom* was the multiplayer aspect of the game, and the deathmatch mode, which allowed players to battle it out against each other in a gameplay environment.

The *Quake* series amped up the multiplayer aspect, so much so that *Quake 3* was released with a very rudimentary single player campaign with heavy focus on multiplayer gameplay. The *Quake 3* engine had a very modular approach, and is one of the most versatile and beautiful game engines around. Modders could change individual characters, or avatars, and change the way they look while playing a match. Modders could change the maps, or create new ones, as an arena in which a deathmatch or capture the flag game is conducted. Modders could also change the underlying physics of the game, the damage of the weapons, the timing of item

spawning, the spawn points themselves, as well as add in entirely new weapons and items to the existing mix. While the modification of *Doom* was restricted to making the game look different, as in say *Doom* with characters from the Batman universe, or from the Alien series of films. *Quake* took the concept of modding way beyond such cosmetic changes, bringing to light a number of nuances of multiplayer gameplay as fundamental considerations during the design of any such game (see box).

OSP (Orange Smoothie Productions) and CPMA (Challenge ProMode Arena)



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are two popular *Quake 3* mods that were played more often than *Quake 3* itself, referred to as *Vanilla Quake*, as the original, unmodded version. These were changes in the physics of the game, in the weapons balance (see box) and a set of map packs which were changed for better gameplay. A typical online *Quake 3* game consisted of custom avatars being played on a custom map using modded physics, all of which were compatible with each other. Sometimes, mods were made to specifically go against the tide of many of the design considerations of other mods. While most *Quake 3* mods focused on easier to navigate

MODIFICATION PRINCIPLES

Spawn Points – Spawn Points in a multiplayer environment are almost of paramount importance. Spawn points are locations on a map, where the player, items or weapons 'appear' at frequent intervals during gameplay. The positioning of the spawn points are important for balanced gameplay.

Weapons Balance – There are a number of weapons in any multi-player game. While designing these weapons, an important consideration should be that any player with a particular weapon does not have an outrageous advantage over another player with a different weapon. The firing time between two consecutive shots, the range the weapon, the availability of ammunition, and the splash damage caused by the weapon are all used to even out the odds of two players using different weapons. Some weapons are known to be tremendously powerful (eg. the BFG and variants) but these are balanced out when amateurish use is as likely to kill the player who is using the weapon as the player who is getting fired at. Additionally, the spawn points of the weapons and ammunition in any single map, are positioned in such a way that the players have a more or less even chance of getting hold of any weapon. The time taken for

ammunition or weapons to re-spawn are also carefully timed after a lot of test gameplay and thought. Weapons Balance also has to be co-ordinated with the Items Balance to equalize the overall gameplay for all the players involved.

Items Balance – Items in the game are things that players can pick up while playing a game. These include the medkits, the armour, power-ups, weapons and ammunition. The spawn points of these items, their positions relative to the spawn points of players and the re-spawn timings of these items have to be carefully planned so that no single player has an inherent advantage while playing a game. Consider a simplistic scenario involving just two players in a deathmatch with two power-ups and no other items except the weapons that they spawn with. One power-up gives a player full armour, and the other power up gives a player full health. Now, the map and the spawn points should be designed in such a way, that no player should have access to both power-ups. Logically, each player should get one of the two power-ups before they continue trying to dismember the other. The same applies for other kinds of games as well, for example the locations of items and creeps in an RTS game.

Technology Tree Balance – In RPG and RTS games, which involves controlling a large number of units which evolve over the gametime, growing in terms of the in-game technology of the races being played, the races often have a unique growth path. Important landmarks in the growth of a race is called a level, and two races at the same level should be evenly matched in a hypothetical perfect game. No race should have an overpowering and obvious advantage over any other race. For RTS games, finding this balance is a very difficult and long process. The amount of damage that a particular unit can do to another unit or building, and the amount of damage that a particular unit can take under attack can be tweaked very slightly, but drastically change the over-all balance of the game. For this reason, most RTS games frequently release patches to continuously even out the balance of the technology growth, so that amongst equally competent players, the races that are used are equally matched.

AI – Sometimes a game can be improved vastly by just tweaking the artificial intelligence of the in-game characters or bots. Many mods for popular games focus on just creating a better AI, and nothing else.